Energy Management of a Renewable-Powered Alkaline Electrolyzer System: A Comparative Study of Nonlinear and Linear Optimization Approaches under Uncertainty

Loukas Kyriakidis[1], Jonaed Bin Mustafa Kamal [1], Saskia Bublitz[1], Bogdan Dorneanu[2], Harvey Arellano-Garcia[2]

[1]German Aerospace Center, Institute of Low-Carbon Industrial Processes, Cottbus, Germany

[2]Brandenburg University of Technology, Chair of Process and Plant Technology, Cottbus, Germany

Abstract

Energy management plays a crucial role in achieving efficient and sustainable operation of industrial energy systems. With the increasing integration of renewable electricity and the growing complexity of hydrogen production networks, effective control strategies are required to minimize operational costs and carbon footprint. However, the uncertain nature of renewable energy sources, such as photovoltaic (PV) power, complicates their accurate forecasting and challenges the optimal energy management of system components. To deal with uncertainties, the rolling-horizon approach (RHA) provides a practical framework for adaptive decision-making by repeatedly solving optimization problems over moving time windows while updating system data in real time. In RHA-based energy management, linear or linearized system models are often employed and optimized by linear methods to reduce computational complexity; however, these simplifications can compromise physical realism and lead to suboptimal decisions. Although RHA can also incorporate local, or global deterministic and stochastic algorithms for nonlinear problems, such approaches frequently suffer from high computational effort, slow convergence, local optima, and difficulty in ensuring constraint satisfaction in large-scale nonlinear systems. To overcome these limitations, this work employs the novel hybrid optimization method BO-IPOPT—a combination of Bayesian Optimization (BO) for global exploration and the Interior Point OPTimizer (IPOPT) for rapid local refinement. The BO-IPOPT method outperformed state-of-the-art approaches in previous energy management studies in accuracy and robustness by achieving lower operational costs at the same CPU time. Here, the hybrid optimizer is compared with other nonlinear optimizers as well as Gurobi for a linearized mixed-integer (MILP) formulation of the system. The study focuses on an industrial hydrogen production system comprising a 30 MW alkaline electrolyzer (AEL), PV generation, grid electricity, an organic Rankine cycle (ORC) unit for waste heat recovery, and high-pressure hydrogen storage. This work specifically examines whether a MILP formulation, often used in the literature, provides a sufficiently accurate simplification for this system, or whether the inherent nonlinearities are significant and must be captured for reliable and cost-effective energy management of renewable hydrogen production. The study also investigates the use of the linearized MILP solution as an initialization (warm start) for the nonlinear optimization to improve convergence and computational efficiency. Finally, the influence of key parameters on the performance of the energy management system is analyzed, including the optimizer's CPU runtime at each RHA iteration, to assess which method performs best under tight computational constraints, as well as uncertainties in PV generation and waste heat availability. The effect of the optimization horizon length is also considered, as longer horizons enable better planning and potentially lower costs but reduce forecast accuracy, highlighting an important trade-off that must be carefully managed.

Keywords: Energy management; Rolling horizon approach; BO-IPOPT; Gurobi; Renewable hydrogen system